

Circulating Biomarkers of Neurodegeneration and All-Cause and Dementia-Specific Mortality

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Abstract

Background and Objectives

Blood-based biomarkers offer a widely available, scalable, and noninvasive method to study neurodegeneration. However, the association between blood-based biomarkers of neurodegeneration and long-term risk of mortality, as well as dementia-specific mortality in a racially diverse cohort, remains understudied. The goal of this study was to determine whether baseline biomarkers of neurodegeneration are associated with long-term risk of all-cause and dementia-specific mortality in a biracial cohort.

Methods

The REasons for Geographic and Racial Differences in Stroke cohort study enrolled 30,239 Black and White participants across the continental United States from 2003 to 2007, with ongoing follow-up. Plasma neurofilament light chain (NfL), total tau, glial fibrillary acidic protein (GFAP), and ubiquitin carboxy-terminal hydrolase L1 (UCH-L1) were measured in baseline plasma from a random sample of participants. All-cause mortality, dementia-specific mortality, cardiovascular-specific mortality, and other causes of death were adjudicated and classified using medical records, the Social Security Death Index, and the National Death Index. Cause-specific Cox regression models accounting for competing risks were used to calculate hazard ratios (HRs) of outcomes for each biomarker separately.

Results

A total of 917 participants had a mean baseline age of 67.4 years (SD 12.1), 49.4% were female, and 48.6% self-identified as Black. With a mean follow-up of 11.1 (SD 5.7) years, 51.0% (477/935) of participants died and 9.2% experienced dementia-specific mortality (86/935). No associations were observed for total tau. In fully adjusted models for other biomarkers, HRs of all-cause mortality per standard deviation increments were 1.93 (95% CI 1.48–2.52) for GFAP, 1.90 (95% CI 1.55–2.32) for NfL, and 1.23 (95% CI 1.09–1.37) for UCH-L1. Furthermore, GFAP (HR 5.66, 95% CI 2.91–11.00) and NfL (HR 2.72, 95% CI 1.57–4.71) were associated with dementia-specific mortality in fully adjusted models. GFAP (HR 2.06, 95% CI 1.22–3.49) and NfL (HR 2.16, 95% CI 1.66–2.81) were also associated with cardiovascular-specific mortality in fully adjusted models.

Discussion

Plasma biomarkers of neurodegeneration, particularly GFAP and NfL, were associated with increased risk of all-cause, dementia-specific, and cardiovascular-specific mortality in a biracial cohort. These associations should be considered when assessing links between these biomarkers and other outcomes, as well as when used in clinical practice.

MORE ONLINE

Supplementary Material

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Glossary

AD = Alzheimer disease; **CATI** = computer-assisted telephone interview; **CES-D-4** = Center for Epidemiological Studies-Depression Scale; **CHAP** = Chicago Health and Aging Project; **CHS** = Cardiovascular Health Study; **GFAP** = glial fibrillary acidic protein; **HANDLS** = Healthy Aging in Neighborhoods of Diversity Across the Life Span; **HR** = hazard ratio; **ICD-10** = International Classification of Diseases, 10th Revision; **IQR** = interquartile range; **NDI** = National Death Index; **NfL** = neurofilament light chain; **REGARDS** = REasons for Geographic and Racial Differences in Stroke; **UCH-L1** = ubiquitin carboxy-terminal hydrolase L1.

Introduction

The number of people living with cognitive impairment and dementia is expected to double by 2050.¹ Alzheimer disease (AD) and dementia deaths have increased over the past 20 years and were the fifth leading cause of death among individuals aged 65 and older in 2021.² Appropriate risk stratification is pertinent to guide diagnostic testing, prognosis, and treatment. This is especially relevant for AD and dementia, given the significant sex and racial differences in causes of death.²

Dementia syndromes are characterized by brain cell death and neurodegeneration. Neurodegeneration is now part of the research criteria for AD and correlates with disease progression and disease staging across a variety of dementia syndromes.^{3,4} Blood-based biomarkers offer a widely available, scalable, and minimally invasive alternative to study neurodegeneration.⁵ Recent research has identified neurofilament light chain (NfL), total tau, glial fibrillary acidic protein (GFAP), and ubiquitin carboxy-terminal hydrolase L1 (UCH-L1) as emerging biomarkers of neurodegeneration.⁶ Thus, we studied the relationships of circulating concentrations of NfL, total tau, GFAP, and UCH-L1 with all-cause mortality and dementia-specific mortality in the biracial longitudinal cohort study, the REasons for Geographic and Racial Differences in Stroke (REGARDS).

Methods

REGARDS Cohort

The REGARDS study is a national, prospective cohort study examining stroke and cognitive function in Black and White adults aged 45 years and older enrolled from 2003 to 2007.⁷ In brief, the cohort was recruited using an initial mailing to a sample of the general population, followed by a telephone call to assess eligibility and interest in participation.⁷ For those providing verbal consent, a computer-assisted telephone interview (CATI) was used to obtain demographics, medical history, questionnaire data (including the Center for Epidemiological Studies-Depression Scale [CES-D-4]), and cognitive testing, as detailed below. The initial CATI was followed within a few weeks by an in-home examination conducted by a trained healthcare professional to collect blood and urine samples, ECG, blood pressure, height, weight, and medications.⁷⁻⁹ The goal was to recruit

approximately half of the participants from the Southeastern United States in an area known as the Stroke Belt (Louisiana, Arkansas, Mississippi, Alabama, Tennessee, Georgia, North Carolina, and South Carolina) and the remainder from the rest of the continental United States.¹⁰ To adequately power the study to examine racial and sex differences in stroke and cognition, REGARDS also sought to balance the cohort by sex and race.

Blood samples from the in-home visits were centrifuged, serum or plasma was separated, and samples were shipped overnight to the Laboratory for Clinical Biochemistry Research at the University of Vermont, the central laboratory for REGARDS, where they were recentrifuged and analyzed or stored at -80°C , as previously described.⁹

Participants in This Study

For this study, we included a random sample of REGARDS participants selected for a case-cohort study of risk factors of stroke and other outcomes.¹¹ The random sample was stratified by age (20% aged 45–54 years, 20% 55–64 years, 25% 65–74 years, 25% 75–84 years, and 10% ≥ 85 years), race (50% Black, 50% White), and sex (50% male, 50% female) to maximize power to detect age and racial differences.^{11,12} From the random sample, we included both participants with and without baseline cognitive impairment.

Definitions

Age, race, sex, region (Stroke Belt vs non-Stroke Belt), level of education (>12 years of education vs ≤ 12 years of education), income (refused, $<\$20,000$, $\$20,000$ – $\$34,000$, $\$35,000$ – $\$74,000$, and $\$75,000$ or above annually), history of stroke, current smoking, alcohol use, and exercise level were self-reported. Race was self-reported using US Census criteria for race. Smoking was self-reported as never, current, or past smoker. Alcohol was classified as heavy use (>7 drinks per week for women, >14 drinks per week for men), moderate use (1–7 drinks per week for women, 1–14 drinks per week for men), and none. Exercise was self-reported as ≥ 4 times per week, 1–3 times per week, and none. History of heart disease was self-reported as heart attack, coronary artery bypass grafting, bypass, angioplasty, or stenting, or evidence of myocardial infarction on ECG. Participants reporting the use of oral glycemic medication or insulin, with fasting glucose >125 mg/dL or nonfasting levels ≥ 200 mg/dL, were considered diabetic at baseline. Hypertension was defined as systolic blood pressure ≥ 130 mm Hg, diastolic blood pressure ≥ 80 mm Hg, or self-reported current medication used to control blood pressure. A

participant was considered to have dyslipidemia if they reported current use of lipid medications, cholesterol ≥ 240 mg/dL, low-density lipoprotein ≥ 160 mg/dL, or high-density lipoprotein ≤ 40.0 . Depressive symptoms were defined as score of 4 or greater on the CES-D-4.^{13,14} Antidepressant use was reported based on the baseline medication inventory. Antidepressant use was included as a covariate because it is often used as a surrogate for depression in observational research. Depression is associated with increased all-cause mortality,¹⁵ dementia-specific mortality,¹⁶ and cardiovascular mortality.¹⁵ Incident stroke was defined as meeting World Health Organization stroke definition or the REGARDS clinical stroke definition after retrieving medical records related to the stroke, which were adjudicated by a panel of stroke experts.^{17,18}

Biomarker Measurements

Baseline EDTA plasma was retrieved from storage, and NfL, total tau, GFAP, and UCHL1 were measured using the ultrasensitive single-molecule array (Simoa) Human Neurology 4-Plex A assay (Quanterix; Cat. No. 102153) and Quanterix HD-1 Analyzer at the REGARDS central laboratory. Measurements were performed blind to participant characteristics.

Mortality Outcomes

During follow-up calls, if a death was reported, death certificates and any medical records for the month preceding death were collected for classification of cause of death. For individuals lost to follow-up, identifiers were sent to the Social Security Death Index and the National Death Index (NDI) to ascertain deaths. All-cause mortality was defined as death due to any cause. Dementia-specific mortality was defined as death due to dementia or AD, including the ICD-10 codes F00–F03, G30, G31.0–G31.1, and R54 on the NDI. Cardiovascular mortality was defined as death due to myocardial infarction, stroke, sudden death, heart failure, pulmonary embolism, other cardiac causes of death, and other noncardiac cardiovascular disease–related deaths. Cardiovascular mortality was defined by adjudication through medical record review in REGARDS. Mortality from other causes was defined as death not related to dementia-specific or cardiovascular-specific mortality.

Statistical Analysis

Analyses were conducted using SAS, version 9.4 (SAS Institute, Cary, NC). Baseline characteristics were tabulated by tertiles of NfL, total tau, GFAP, and UCH-L1. Analyses used tertiles and continuous log-transformed neurodegeneration biomarkers expressed per SD increment of the distributions, and each biomarker was considered separately. To examine the risk of all-cause mortality and dementia-specific mortality while accounting for competing risks, cause-specific Cox models were used separately for each biomarker. In addition to the main outcome of all-cause mortality, the analysis also considered the competing risk of loss to follow-up (participants inactive in REGARDS). For dementia-specific mortality, competing risks also included cardiovascular mortality and mortality from all other causes (e.g., cancer or renal

disease). Incrementally adjusted models were used; the crude model was unadjusted. Model 1 adjusted for age, sex, race, region, education, and income. Model 2 further adjusted for hypertension, dyslipidemia, and diabetes mellitus. Model 3 added a history of cognitive impairment, prebaseline stroke, and incident stroke as time-varying covariates. Finally, model 4 further adjusted for exercise level, smoking status, alcohol use, antidepressant use, and depressive symptoms. Interaction terms for race, sex, and age were tested in the fully adjusted model, with a *p* value for interaction < 0.05 considered statistically significant. In addition to the competing risk models, we used cumulative incidence functions to estimate and visualize the incidence of each of the competing events.¹⁹

Standard Protocol Approvals, Registrations, and Patient Consents

REGARDS is an ongoing study conducted according to guidelines in the Declaration of Helsinki, and all procedures involving human subjects were approved by the relevant institutional review board at participating universities. Verbal and written informed consent was obtained from all participants.

Data Availability

This study used data from the REGARDS cohort. To abide by its obligations with NIH/National Institute of Neurological Disorders and Stroke and the Institutional Review Board of the University of Alabama at Birmingham, REGARDS facilitates data sharing through formal data use agreements. Any investigator is welcome to access the REGARDS data through this process. Requests for data access may be sent to regardsadmin@uab.edu.

Results

The REGARDS cohort random sample included 1,104 participants; 88 were excluded because of missing biomarkers, and 99 were excluded because of missing covariates, resulting in a sample size of 917 (Figure 1). Cohort characteristics are given in Table 1; compared with the overall cohort, participants with all-cause mortality were more likely to be male, were older, earned $< \$75,000$ per year, had a history of heart disease and stroke, and had baseline cognitive impairment.

Figure 1 Flowchart of Case Enrollment

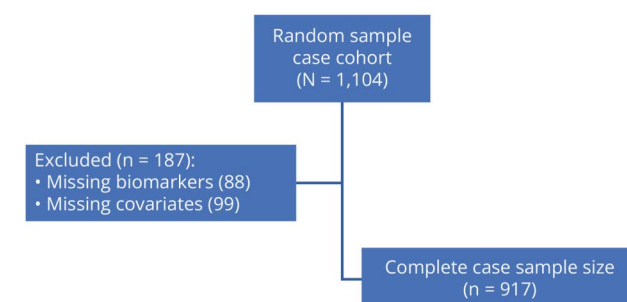


Table 1 Baseline Characteristics Overall and by Mortality Type

	Overall (N = 917)	All-cause mortality (N = 459)	Dementia mortality (N = 83)
Female, n (%)	458 (50.0)	212 (46.2)	52 (62.7)
Black, n (%)	444 (48.4)	227 (49.5)	40 (48.2)
Age, y, mean $\pm$ SD	67.2 $\pm$ 12.1	74.2 $\pm$ 10.1	78.9 $\pm$ 8.1
Stroke Belt, n (%)	482 (52.6)	240 (52.3)	40 (48.2)
>12-y education, n (%) ^a	583 (63.6)	269 (58.6)	51 (61.5)
Income, n (%)			
<\$20K	161 (17.6)	108 (23.5)	14 (16.9)
\$20K–\$34K	223 (24.3)	127 (27.7)	22 (26.5)
\$35K–\$74K	267 (29.1)	124 (27.0)	23 (27.7)
$\geq$ \$75K	137 (14.9)	31 (6.8)	5 (6.0)
Refused	129 (14.1)	69 (15.0)	19 (22.9)
History of hypertension, n (%)	682 (74.4)	366 (79.7)	70 (84.3)
History of dyslipidemia, n (%)	528 (57.6)	281 (61.2)	44 (53.0)
History of heart disease, n (%)	163 (17.8)	117 (25.5)	21 (25.3)
History of diabetes mellitus, n (%)	183 (20.0)	115 (25.1)	17 (20.5)
Prevalent stroke, n (%)	63 (6.9)	54 (11.8)	8 (9.6)
Incident stroke, n (%)	70 (7.6)	45 (9.8)	4 (4.8)
Baseline cognitive impairment, n (%)	97 (10.6)	72 (15.7)	18 (21.7)
Lifestyle factors, n (%)			
Smoking			
Current	127 (13.9)	63 (13.7)	6 (7.2)
Never	441 (48.1)	198 (43.1)	41 (49.4)
Past	349 (38.1)	198 (43.1)	36 (43.4)
Alcohol use			
Heavy	35 (3.8)	16 (3.5)	4 (4.8)
Moderate	298 (32.5)	128 (27.9)	25 (30.1)
None	584 (63.7)	315 (68.6)	54 (65.1)
Exercise			
1–3 times per week	328 (35.8)	151 (32.9)	26 (31.3)
$\geq$ 4 times per week	265 (28.9)	123 (26.8)	25 (30.1)
None	324 (35.3)	185 (40.3)	32 (39.4)
Antidepressant medication use	109 (11.9)	54 (11.8)	9 (10.8)
Depressive symptoms	107 (11.7)	60 (13.1)	12 (14.5)

^a Some college or college graduate.

Participants with dementia-specific mortality were more likely to be female, were older, earned <\$75,000 per year, had a history of heart disease, had baseline cognitive impairment, and had a lower exercise frequency.

From baseline to 2019, 459 participants died, 83 with dementia mortality and 110 with cardiovascular mortality. The median follow-up time was 9.5 years (interquartile range [IQR] 5.6–12.7 years) for those who died, 8.6 years (IQR

4.0–13.9 years) for those lost to follow-up, and 13.1 years (IQR 12.1–14.1 years) for those still in follow-up in 2019. The tertiles for GFAP were as follows: T1 <109.8, T2 109.8–188.4, and T3 ≥188.4 ng/mL. Tertiles of NfL were as follows: T1 <10.4, T2 10.4–17.9, and T3 ≥17.9 pg/mL. Total tau tertiles were categorized as T1 <1.37, T2 1.37–2.13, and T3 ≥2.14 pg/mL. Finally, UCH-L1 tertiles were categorized as T1 <51.7, T2 51.7–86.7, and T3 ≥86.7 ng/L.

eTable 1 lists participant characteristics by tertile of neurodegenerative biomarkers. Those with higher GFAP were female; were older; had income <\$75,000; were hypertensive; had a history of heart disease, baseline stroke, and incident stroke; abstained from alcohol; and did not exercise. Participants with higher NfL were more likely to be older; earn less than <\$75,000 per year; have hypertension, heart disease, stroke, and incident stroke; be former smokers; abstain from alcohol; and not exercise. Higher total tau was associated with female sex, diabetes mellitus, and lack of exercise. Higher UCH-L1 was associated with female sex, White race, older age, income <\$75,000, heart disease, incident stroke, and lack of exercise.

Event frequencies and survival averages for the study outcomes by biomarker levels are given in eTable 2.

Associations With All-Cause Mortality

Cumulative incidence plots for time to all-cause mortality for each biomarker are shown in Figure 2. In Table 2, GFAP, NfL, and UCH-L1, but not total tau, were associated with all-cause mortality in the fully adjusted model. eTable 3 demonstrates biomarker associations with all-cause mortality in each model. Total tau had a more modest association across the models, but the association was attenuated after adjusting for lifestyle covariates in model 4. For GFAP, in the fully adjusted model, the highest tertile had a cause-specific hazard ratio (HR) of 2.20 (95% CI 1.46–3.29) compared with the lowest tertile of GFAP, and in the fully adjusted model treating GFAP levels continuously, the HR was 1.88 (95% CI 1.43–2.46) per SD increment. Similar results were seen in the fully adjusted model for the highest vs lowest tertile of NfL (HR 2.87, 95% CI 1.82–4.54) and UCH-L1 (HR 1.40, 95% CI 1.09–1.79), as well as the HR per SD increment of NfL (HR 1.89, 95% CI 1.54–2.32) and UCH-L1 (HR 1.22, 95% CI 1.08–1.37).

The association of GFAP tertiles and continuous GFAP (with all-cause mortality differed significantly by race with larger associations among White than Black participants (respective *p* for interaction 0.02 and 0.002). For Black participants in the fully adjusted model, the highest tertile of GFAP had a cause-specific HR of 1.21 (95% CI 0.79–1.87) compared with the lowest tertile, and in the fully adjusted model treating GFAP levels continuously, the HR was 1.18 (95% CI 0.92–1.51) per SD increment. For White participants in the fully adjusted model, the highest tertile of GFAP had a cause-specific HR of 2.12 (95% CI 1.33–3.38) compared with the lowest tertile, and in the fully adjusted model treating GFAP levels

continuously, the HR was 1.84 (95% CI 1.34–2.53) per SD increment. There was also a significant difference by race with a similar pattern for NfL tertiles (*p* interaction = 0.02) and continuous NfL (*p* interaction = 0.006). For Black participants in the fully adjusted model, the highest tertile of NfL had a cause-specific HR of 1.51 (95% CI 0.95–2.39) compared with the lowest tertile, and in the fully adjusted model treating NfL levels continuously, the HR was 1.50 (95% CI 1.22–1.85) per SD increment. For White participants in the fully adjusted model, the highest tertile of NfL had a cause-specific HR of 2.56 (95% CI 1.50–4.37) compared with the lowest tertile, and in the fully adjusted model treating NfL levels continuously, the HR was 1.86 (95% CI 1.47–2.37) per SD increment.

Associations With Dementia-Specific Mortality

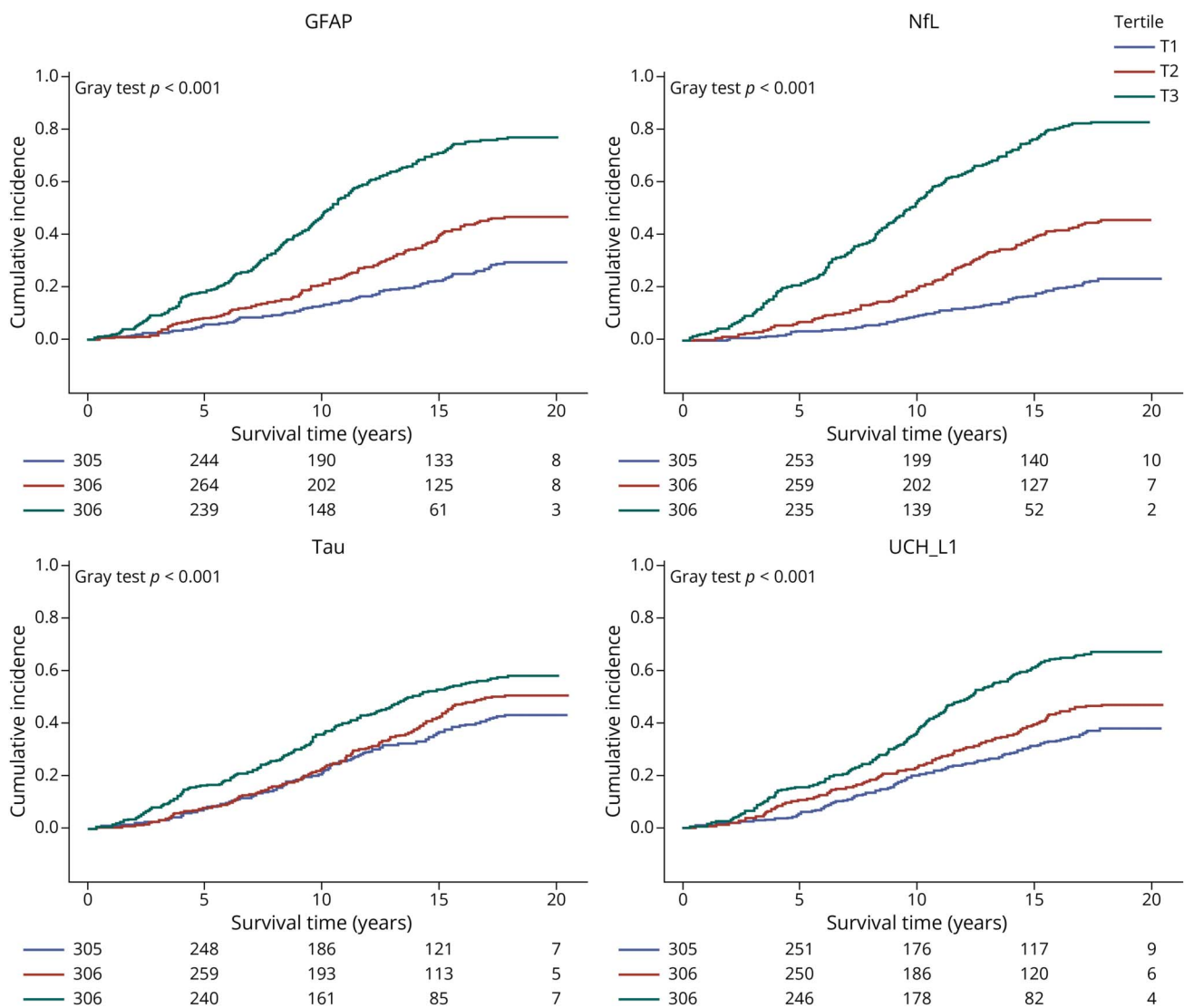
Cumulative incidence plots for time to dementia-specific mortality for each biomarker are shown in Figure 3. As given in Table 2, associations of each SD increment of GFAP and NfL with dementia-specific mortality were strong in the fully adjusted model. The highest tertile of GFAP was associated with a cause-specific HR of 5.51 (95% CI 1.85–16.39), and the HR per SD increment was 5.66 (95% CI 2.91–11.00). Similar associations were observed with NfL: the highest tertile had a cause-specific HR of 2.82 (95% CI 1.09–7.31), with a HR per SD increment of 2.72 (95% CI 1.57–4.71) in the fully adjusted model. Across incrementally adjusted models, associations were partly attenuated after accounting for socio-demographics (model 1) for GFAP and NfL, but no further attenuation was observed (eTable 4). Comparing the highest with the lowest tertile of total tau, the fully adjusted HR was 3.19 (95% CI 1.34–7.62) for dementia-specific mortality (eTable 4). However, no association was observed when total tau was treated continuously (HR per SD 1.22, 95% CI 0.84–1.76). UCH-L1 was not associated with dementia-specific mortality in the fully adjusted models assessing UCH-L1 in tertiles or continuously.

There was a significant difference in the association of continuous GFAP with dementia-specific mortality by race (*p* for interaction = 0.01). For Black participants in the fully adjusted model treating GFAP levels continuously, the HR was 1.68 (95% CI 0.83–3.39) per SD increment. For White participants in the fully adjusted model treating GFAP levels continuously, the HR was 8.92 (95% CI 4.18–19.07) per SD increment. There was also a difference in the association of continuous NfL with dementia-specific mortality by race (*p* interaction = 0.008). For Black participants in the fully adjusted model treating NfL levels continuously, the HR was 1.11 (95% CI 0.62–1.99) per SD increment. For White participants in the fully adjusted model treating NfL levels continuously, the HR was 3.66 (95% CI 1.97–6.81) per SD increment.

Associations With Cardiovascular-Specific Mortality

Cardiovascular-specific mortality for each biomarker was also examined, with cumulative incidence plots shown in Figure 4 and HRs listed in Table 2 and eTable 5. Here, NfL was the

Figure 2 Cumulative Incidence Plots for Time to All-Cause Mortality by Biomarker



GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; UCH-L1 = ubiquitin carboxy-terminal hydrolase L1.

most robust marker, with a fully adjusted HR of 5.41 for the highest tertile vs lowest tertile (95% CI 2.33–12.60) and a HR of 2.27 per SD increment (95% CI 1.72–2.98). For GFAP, the HR per SD increment was 2.13 (95% CI 1.25–3.62), and

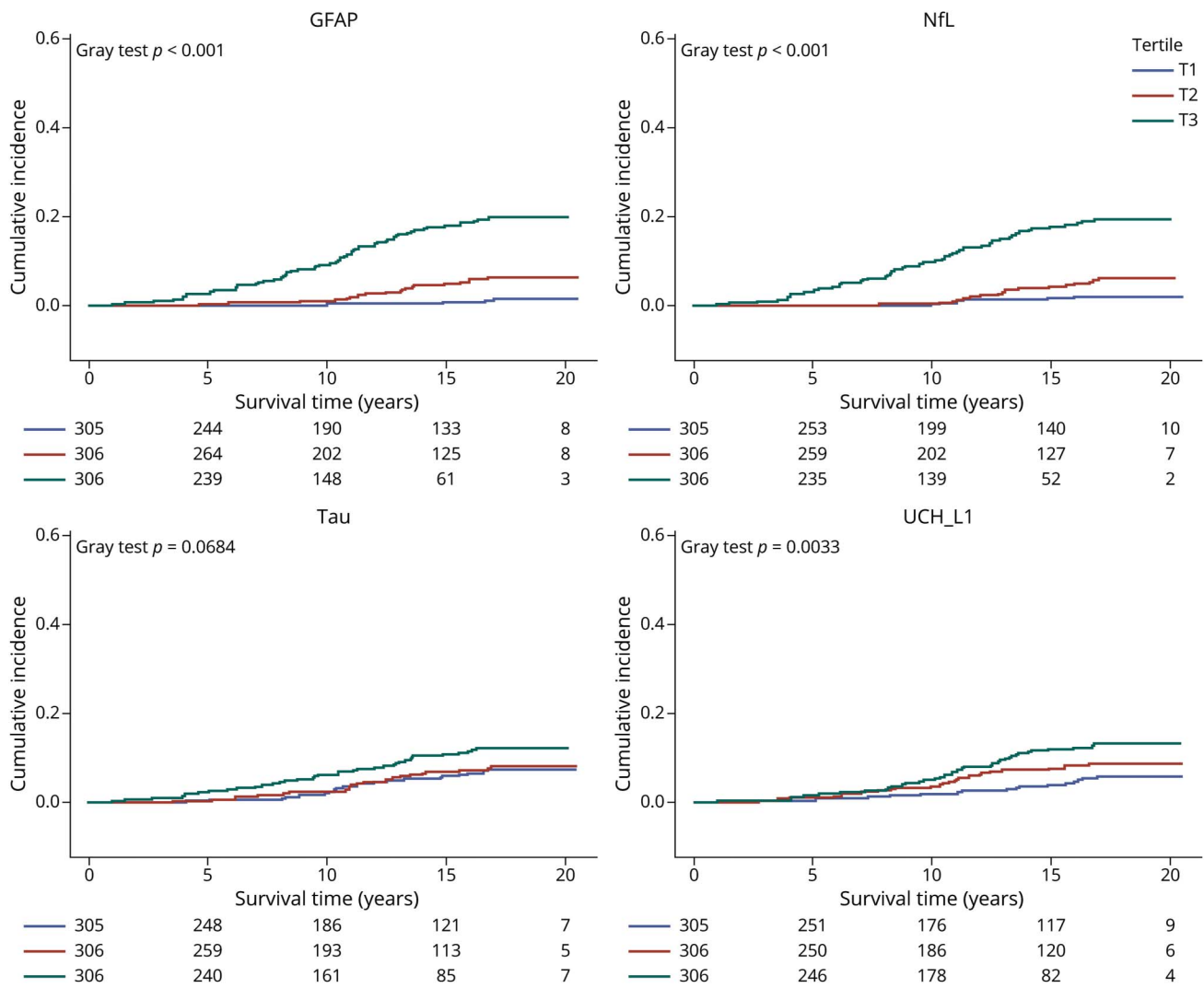
while the HR was increased, the association between of the top vs bottom tertile was not statistically significant (HR 1.59, 95% CI 0.83–3.05). A similar pattern was observed for UCH-L1, which had lower HRs. Across incrementally adjusted

Table 2 Cause-Specific Hazard Ratio of Mortality for a 1 SD Increment of Each Plasma Biomarker

Biomarker	All-cause mortality (95% CI)	Dementia-specific mortality (95% CI)	Cardiovascular-specific mortality (95% CI)
GFAP	1.88 (1.43–2.46)	5.47 (2.77–10.80)	2.13 (1.25–3.62)
NfL	1.89 (1.54–2.32)	2.69 (1.53–4.73)	2.27 (1.72–2.98)
UCH-L1	1.22 (1.08–1.37)	1.14 (0.84–1.55)	1.29 (1.01–1.64)
Tau	0.35 (0.11–1.10)	1.22 (0.84–1.76)	1.23 (0.87–1.73)

Abbreviations: GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; UCH-L1 = ubiquitin carboxy-terminal hydrolase L1. Adjusted for age; sex; race; region; education (college/vs no college); income; hypertension; dyslipidemia; diabetes mellitus; history of cognitive impairment, stroke, and incident stroke; exercise; smoking status; alcohol use; antidepressant use; and depression.

Figure 3 Cumulative Incidence Plots for Time to Dementia-Specific Mortality by Biomarker



GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; UCH-L1 = ubiquitin carboxy-terminal hydrolase L1.

models, associations were partly attenuated after accounting for sociodemographics (model 1) for GFAP, NfL, and UCH-L1, but no further attenuation was observed (eTable 5). Total tau was not associated with cardiovascular-specific mortality in the fully adjusted models.

Similar to the other race interactions, the association of continuous GFAP with cardiovascular-specific mortality differed by race ($p = 0.004$). For Black participants in the fully adjusted model treating GFAP levels continuously, the HR was 1.02 (95% CI 0.63–1.66) per SD increment. For White participants in the fully adjusted model treating GFAP levels continuously, the HR was 1.73 (95% CI 0.92–3.28) per SD increment.

Associations With Mortality From Other Causes and Loss to Follow-Up

Cause-specific mortality HRs were also calculated for biomarkers and mortality due to other causes (eTable 6). UCH-L1 was associated with mortality from other causes in model 4; the highest

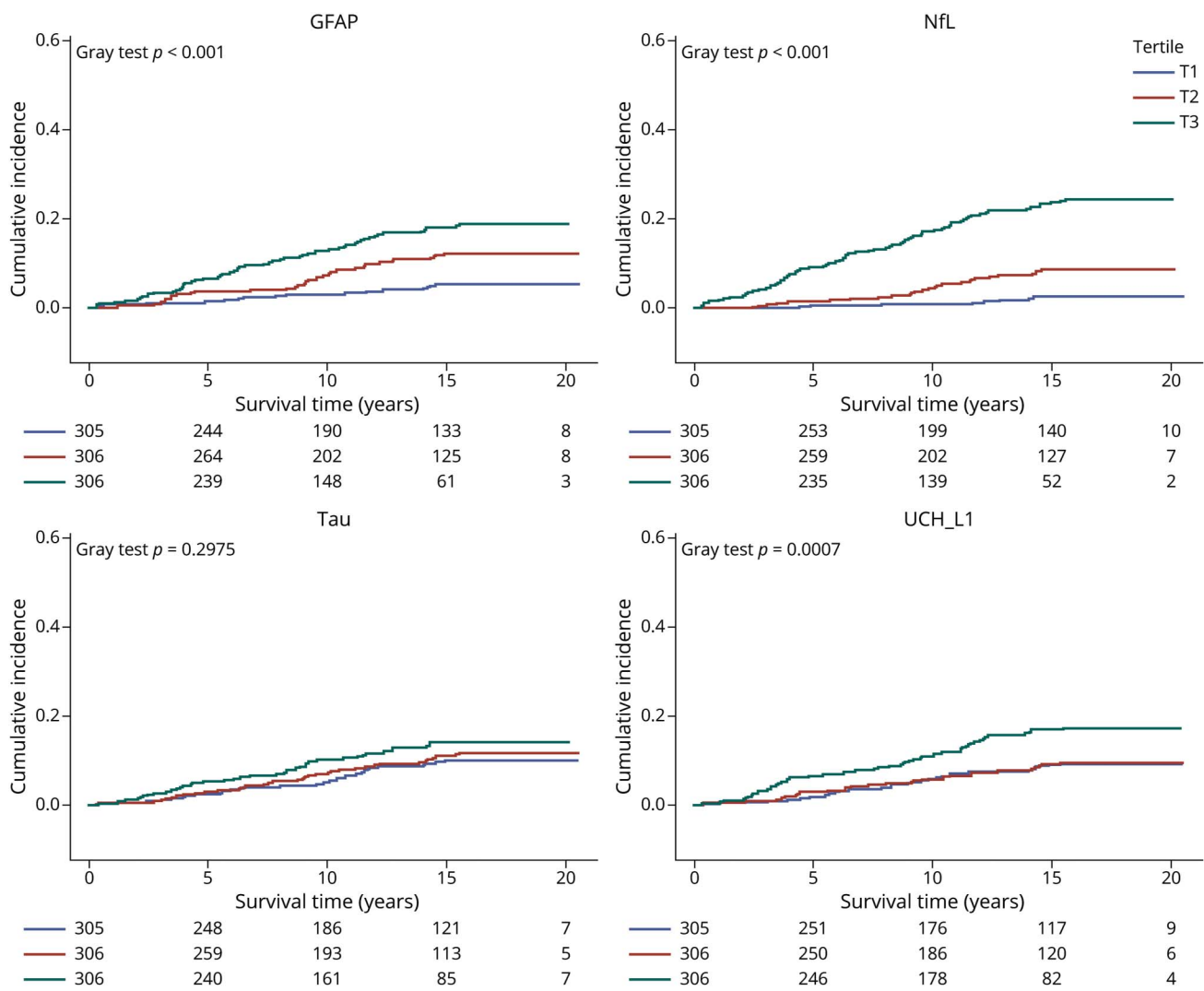
tertile had an HR of 1.50 (95% CI 1.08–2.07) compared with the lowest tertile, and the HR per SD increment was 1.22 (95% CI 1.05–1.42). Similar to other outcomes, these associations were modestly attenuated after accounting for sociodemographic variables (model 1), but not in the later models. Higher NfL was associated with mortality from other causes when treated continuously (HR 1.33, 95% CI 1.08–1.64) but by tertiles. GFAP and total tau were not associated with mortality due to other causes.

Associations of biomarkers with loss to follow-up were also assessed (eTable 7). In the fully adjusted model, only each higher SD GFAP was associated with loss to follow-up; HR per SD increment 3.48 (95% CI 1.15–10.47), with no association by tertiles.

Discussion

In this population-based biracial cohort study of participants with and without baseline cognitive impairment, higher

Figure 4 Cumulative Incidence Plots for Time to Cardiovascular-Specific Mortality for Each Biomarker



GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; UCH-L1 = ubiquitin carboxy-terminal hydrolase L1.

baseline plasma concentrations of GFAP and NfL were associated with greater all-cause, dementia, and cardiovascular mortality over 9.5 years accounting for competing risks. These associations were driven by the results seen in White participants. UCH-L1 was associated with all-cause mortality and mortality from other causes. Total tau was associated with dementia-specific mortality. In general, findings support the lack of specificity of these biomarkers for dementia mortality, although associations of GFAP and NfL with dementia mortality were larger.

Our findings in REGARDS agree with previous population-based studies of participants without dementia, which found that circulating NfL was associated with all-cause mortality.²⁰⁻²⁸ Along with the Healthy Aging in Neighborhoods of Diversity Across the Life Span (HANDLS) study, our study had the largest representation of Black Americans.²⁴ Similar to results from the Cardiovascular Health Study (CHS) (HR for NfL in

the highest compared with the lowest tertile 2.87, 95% CI 1.79–4.61), we found that plasma NfL was associated with dementia-specific mortality (HR 2.82 for the highest tertile compared with the lowest tertile, 95% CI 1.09–7.31).²⁸ The mean age of study participants in REGARDS was younger than in CHS (67 years old vs 77 years old), and REGARDS had a greater representation of Black persons (49% vs 13%), which may increase the generalizability of our findings. Two population-based studies of participants without dementia examined the association between NfL and cardiovascular-specific mortality,^{26,29} both supporting our findings that higher plasma NfL is associated with long-term cardiovascular-specific mortality. One possible explanation is that higher plasma NfL is associated with increased systemic inflammation,³⁰ which is linked to cardiovascular-specific mortality.³¹ Another possible explanation is that many risk factors of neurodegeneration are also cardiovascular risk factors,³² although we adjusted for these.

GFAP has gained attention as a marker of cerebral dysfunction because of its specificity for brain astrocytes, low level of expression in healthy people, and marked elevation in the brains of patients with AD.³³ Our findings that higher levels of GFAP are associated with all-cause mortality (HR 2.45 for top tertile, 95% CI 1.63–3.66) were more robust than those found in 1,327 older participants from the Chicago Health and Aging Project (CHAP), a biracial longitudinal population-based study (HR 1.43 for a fivefold higher concentration, 95% CI 1.08–1.90).²⁹ As in REGARDS, some participants had baseline cognitive impairment in CHAP.^{34,35} We were able to adjust for several dementia risk factors, including demographics, medical history, lifestyle habits, history of cognitive impairment, and prevalent/incident stroke, which were not adjusted for in CHAP. Participants in REGARDS were younger than in CHAP, represented a national sample, and were a random sample of a large cohort (by contrast to the report from CHAP, where participants were selected for factors occurring after baseline). Differences may also be due to higher rates of mortality seen in CHAP, where 59% of participants were deceased at the time of their evaluation. The CHAP study also observed an association between plasma GFAP and cardiovascular mortality.²⁹ For dementia-specific mortality, we found a stronger association with GFAP (HR 5.51, 95% CI 1.85–16.39) than in the CHS (HR 2.76 for the highest tertile compared with the lowest tertile, 95% CI 1.72–4.40), which included 1,712 dementia-free participants, of whom 13% were Black Americans.²⁸

The association between plasma/serum total tau and mortality has been studied much more in neurodegenerative diseases³⁶ or cerebral insults,^{37,38} than in healthy aging populations. In 385 participants from the Memory and Morbidity in Augsburg Elderly study, total tau was associated with all-cause mortality (HR per SD 1.20, 95% CI 1.07–1.35).²⁵ However, when total tau was added to a model containing sociodemographic factors, it did not improve all-cause mortality prediction.²⁵ This aligns with our findings that in an unadjusted model, plasma total tau was associated with all-cause mortality; however, in the fully adjusted models, there was no association. In CHS, in fully adjusted models, the highest tertile of plasma total tau was not associated with dementia-specific mortality (HR 1.03, 95% CI 0.85–1.25) compared with the lowest.²⁸ This contrasts with our findings in which the highest tertile of plasma total tau was strongly associated with dementia-specific mortality. Differences between the 2 studies discussed above may explain the disparate findings. In the CHAP study, after adjusting for age, sex, education, APOE-ε4, body mass index, number of chronic health conditions, global cognition score, and physical functioning score, total tau was not associated with cardiovascular-specific mortality, like results here.

Most of the studies focusing on UCH-L1 and mortality are in traumatic brain injury populations³⁹ or acute injuries.^{40,41} REGARDS does not have data on previous or recent traumatic brain injury so it could not be included as a covariate in

analysis. We are not aware of other studies on the association between plasma UCH-L1 and mortality in an initially healthy population. We observed no association between UCH-L1 and all-cause, dementia-specific, and cardiovascular-specific mortality. However, higher UCH-L1 was associated with mortality from other causes (HR 1.52 for the highest tertile compared to the lowest, 95% CI 1.11–2.08). While other markers of neurodegeneration seem to have exclusively neural or glial cell origins, UCH-L1 has been found in breast,^{42,43} prostate,⁴⁴ and pancreatic cancers.⁴⁵ As these cancers increase mortality and might increase UCH-L1 levels in plasma, this could explain why UCH-L1 was associated with mortality due to other causes in this study. However, REGARDS participants could not have active cancer at baseline, so further research is warranted before any firm conclusions can be drawn.

Our results demonstrated racial differences in the association between plasma biomarkers of neurodegeneration and all-cause, dementia-specific, and cardiovascular-specific mortality. Previous work has demonstrated differential association between plasma biomarkers and other markers of neurodegenerative disease in Black participants and White participants.^{46,47} In addition, plasma biomarkers of neurodegeneration have been shown to have different associations with cognitive impairment/decline⁴⁸ and functional abilities⁴⁹ in Black participants and White participants. The HANDLS study had the largest representation of Black participants in its study population and included race as a covariate in their analysis of sex differences, but it did not specifically investigate differences in the association between plasma biomarkers of neurodegeneration and mortality in Black participants compared with White participants.²³ To our knowledge, ours is the first study to present data on racial differences in the association between mortality and plasma biomarkers of neurodegeneration.

The strengths of this study include its population-based biracial design, high longitudinal retention (97% annually), and long follow-up duration. Use of ultrasensitive immunoassays for baseline plasma GFAP, NfL, total tau, and UCH-L1 with rigorous quality control practices was also a strength. In addition, use of competing risk analysis strengthens interpretation of the findings. The study had several limitations; although the REGARDS cohort has a significant representation of Black participants, it only included 2 racial groups, which may limit its generalizability to a broader population. The cohort sample included here was also relatively small, so some associations might have been missed. The youngest age was 45 years, and the study of even younger people might be helpful to extend the findings. In addition, plasma biomarkers were only measured once, and levels may change over time. There are several limitations to using ICD-10 codes in research, including inconsistent coding practices across health care settings, lack of specificity for rare or complex diagnoses, and potential loss of clinical context from the full medical record, which increases the risk of misclassifying patients.⁵⁰ Another limitation of our research is that cerebral small vessel

disease was not assessed as a confounder; this is relevant because stroke and cardiovascular disease are present in this population. Finally, although many potential confounding factors were included, there may still be unaccounted factors, so caution should be exercised in interpreting the results.

Plasma biomarkers of neurodegeneration, particularly NfL and GFAP, were associated with all-cause mortality, dementia-specific mortality, and cardiovascular mortality in a biracial cohort of over 10 years of follow-up. The markers of brain cell death that were used in our study differ from those that have become commercially available to assess underlying Alzheimer pathologic changes in peripheral blood, notably amyloid beta and hyperphosphorylated tau species. After verification of our findings in additional cohorts, these blood-based biomarkers will be scalable and well suited for population-based studies or assessing clinical prognosis regarding overall and cause-specific mortality.

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Author Contributions

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Disclosure

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